

Agentic AI Architecture

& Microsoft Foundry Agent Service

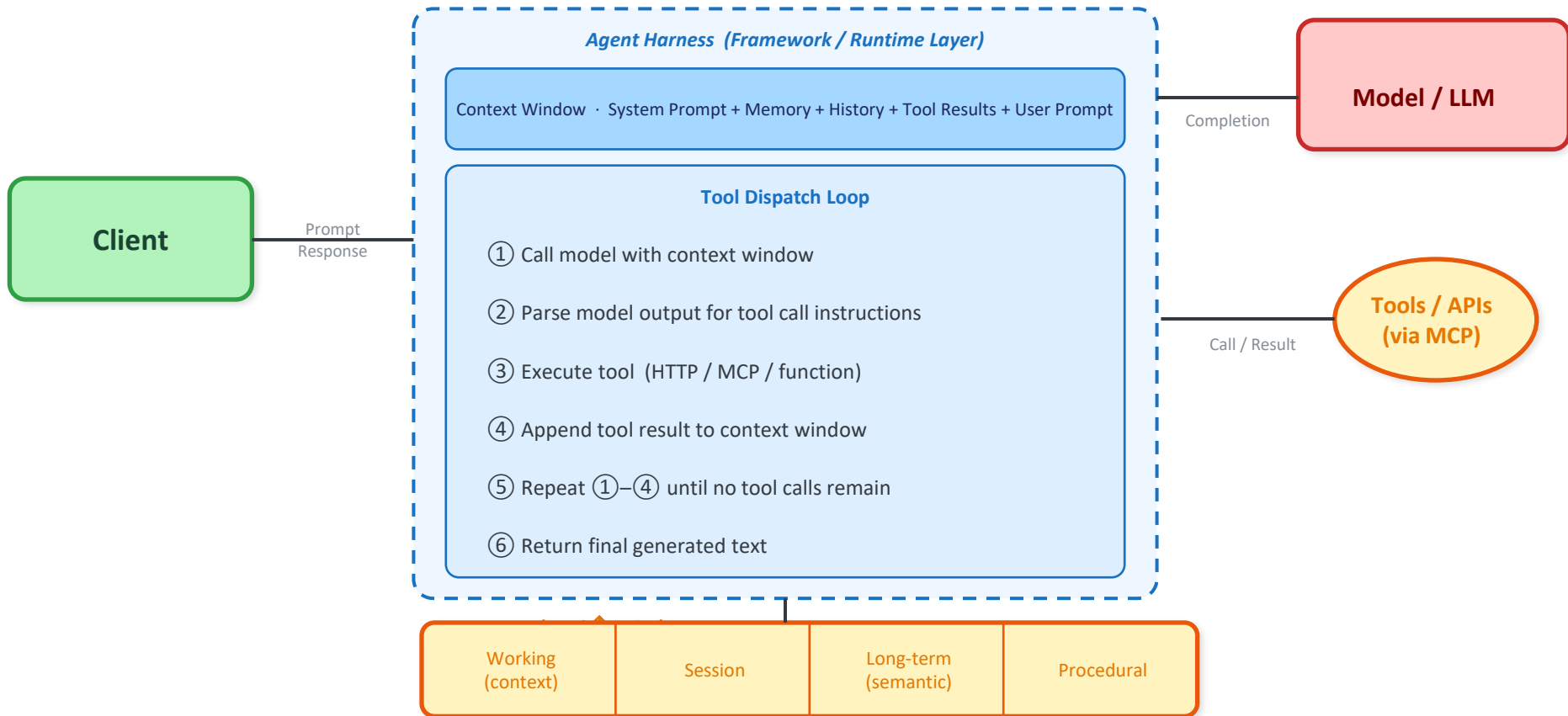
Solving for production

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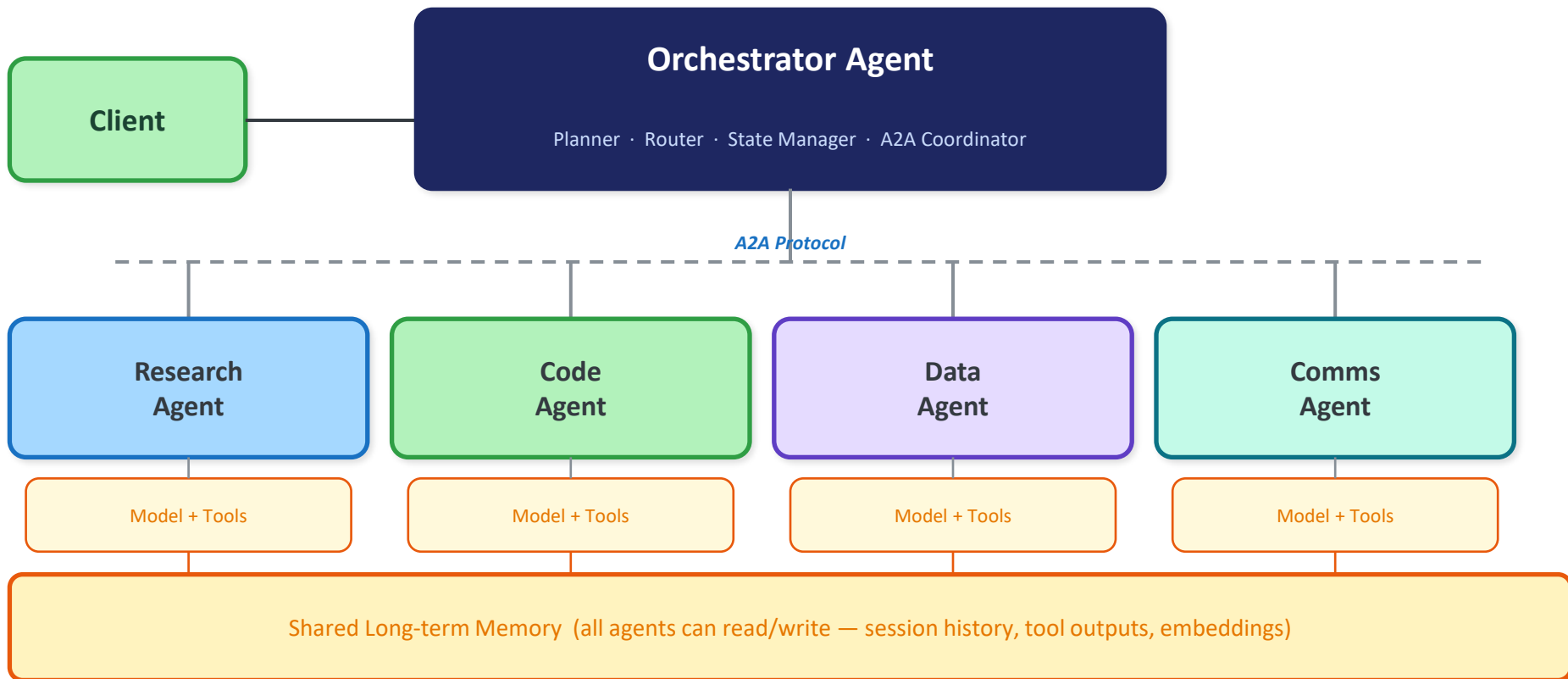
Complete Single-Agent Architecture

The harness owns the loop — the model only generates text



Scaling to Multi-Agent

Orchestrator plans and routes — sub-agents execute in parallel via the A2A protocol



Where Things Break: 8 Failure Surfaces

Every hop in the agentic loop is a potential point of failure

#	Failure Zone	Root Cause	Impact
①	Client → Agent	Network timeout / dropped connection	<i>Request lost, no response returned</i>
②	Context Window	Token limit exceeded (context overflow)	<i>Model truncates; loses earlier context</i>
③	Agent → Model	Model API timeout, throttle (HTTP 429), or outage	<i>Loop stalls; partial or no answer</i>
④	Tool Dispatch	Tool call timeout, auth failure, or 5xx error	<i>Tool result missing; model reasons incorrectly</i>
⑤	Tool → Agent	Network partition or high latency on result return	<i>Loop hangs indefinitely without retry</i>
⑥	Tool Loop	No stop condition → infinite recursion	<i>Runaway cost and compute; no output</i>
⑦	Memory Store	DB unavailable / write failure (Cosmos, Search)	<i>Context lost across sessions; stale data</i>
⑧	Multi-agent / A2A	Sub-agent crash or A2A handoff failure	<i>Orchestrator stalls; partial work discarded</i>

Microsoft Foundry Agent Service

A managed platform that owns the harness, memory, and resiliency — so you don't have to build it

BUILD

- Supports Agent Framework, LangGraph, OpenAI Agents SDK, Anthropic Agent SDK, GitHub Copilot SDK — no rewrites
- Prompt Agents (portal, no code needed)
- Toolbox: 1,400+ MCP-compatible tools
- Centralized tool versioning & promotion

DEPLOY

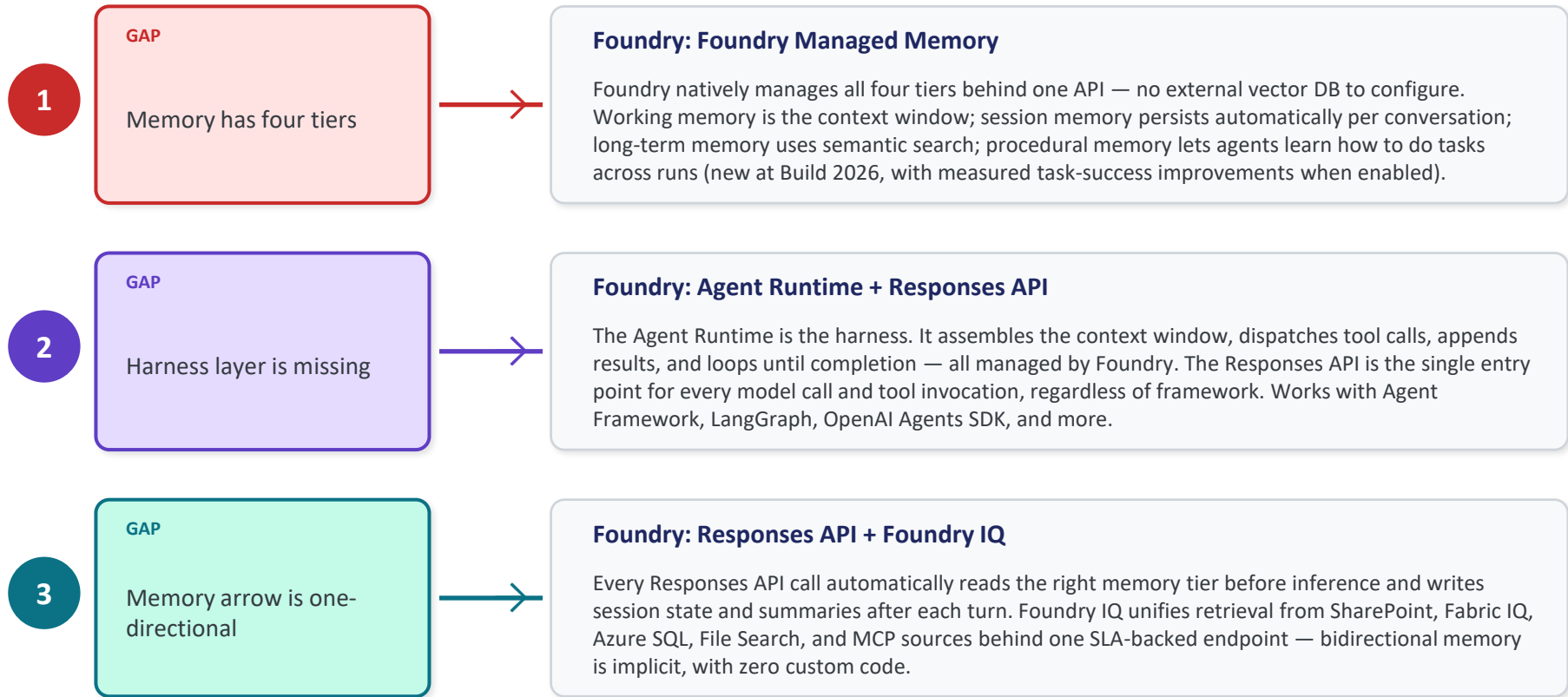
- Hosted Agents — containerized, auto-scaled
- Session-level state persistence
- Long-running agents & scheduled routines
- Dedicated Microsoft Entra identity per agent
- Publish to Teams, M365 Copilot, or API
- Private VNet support with BYO VNet

OPERATE

- Full OTel tracing: every model call, tool hop, and sub-agent handoff
- Evaluations linked back to traces
- Agent Optimizer (auto-improves instructions)
- Zone-redundant HA infrastructure
- Disaster recovery runbooks provided

Foundry Closes the Architecture Gaps

The three additions from Slide 3 — Foundry handles each one natively



Foundry Addresses Every Failure Point

Direct answer for each of the 8 failure zones — from networking to runaway loops

Network timeout (client → agent)

①



Managed Azure load-balanced endpoint; Standard deployments spread requests across regions automatically

Context window overflow

②



Runtime enforces token limits; session + procedural memory offloads history so context stays lean

Model API timeout / HTTP 429 throttle

③



Built-in retry with backoff; multi-region PTU deployments via Foundry model catalog

Tool dispatch failure (auth / 5xx)

④



Managed auth (Entra identity + OBO passthrough); every failed tool call is surfaced in OpenTelemetry trace

Network partition (tool → agent)

⑤



Toolbox SLA-backed retrieval; zone-redundant storage for tool state; configurable timeouts & retries

Infinite tool loop

⑥



Agent Runtime enforces recursion limits and stop conditions; Agent Optimizer flags looping patterns

Memory store unavailable

⑦



Zone-redundant Cosmos DB + Azure AI Search; BYO resources for compliance; session state checkpointed

Sub-agent crash / A2A failure

⑧



Hosted agents run in isolated VM sandboxes with independent scaling; A2A returns structured error responses

Dapr Agents: When You Host on AKS

Dapr Agents is the self-hosted answer — same distributed-systems guarantees, fully on your infrastructure

Dapr Agents (CNCF / v1.0 / open-source) builds on Dapr's distributed-systems runtime — durable workflows, pub/sub messaging, virtual actor model, state management, and 50+ component connectors — giving you an enterprise-grade agent harness on any Kubernetes cluster.

AKS + Dapr Agents (Self-Hosted)

vs

Foundry Hosted Agents (Managed)

Full infra control — runs on any K8s cluster, any cloud or on-prem

Virtual actor model → scale-to-zero; thousands of idle agents at near-zero cost

Durable workflows: checkpointed, resumable after any crash or node failure

Pub/Sub event-driven messaging between agents (Kafka, Redis, Azure SB, etc.)

Vendor-neutral (CNCF); swap LLM, state store, or message bus without code change

Resiliency policies: circuit breakers, retry/backoff, timeouts per component

50+ Dapr component connectors + MCP support for data ingestion

DIY identity, networking, secrets; integrate with existing platform team patterns

Zero infra to manage — Foundry runs compute, scaling, networking

Auto-scale per session volume; per-call billing for prompt agents

Built-in session persistence and long-running agent support

A2A protocol for structured agent-to-agent handoff and routing

Multi-model catalog (GPT-4o, Llama, DeepSeek); swap model without code change

Managed retry, multi-region PTU, zone-redundant HA out of the box

1,400+ MCP tools; Foundry IQ for SharePoint, Fabric, Azure SQL retrieval

Managed Entra identity per agent; OAuth OBO passthrough; private VNet

Choose AKS + Dapr Agents when: multi-cloud/on-prem, existing K8s platform, scale-to-zero budget, or open-source governance required. Choose Foundry Hosted when: Azure-first, Teams/M365 publishing, fastest time to production, or no platform team to manage infra.

From Diagram to Production

1

The baseline agentic loop correctly captures the core flow: bidirectional client communication, system prompt, MCP tool registration, loop back to model, and optional agent handoff.

2

Three additions sharpen it: 4-tier memory, an explicit harness/dispatch layer, and bidirectional memory read/write.

3

Multi-agent adds an orchestrator, A2A routing, and shared long-term memory across specialized sub-agents running in parallel.

4

8 failure surfaces span every hop: network, token overflow, tool errors, memory outages, and runaway loops.

5

Foundry Agent Service = managed production answer. Dapr Agents on AKS = self-hosted open-source alternative. Same guarantees, different operating model.